

COMMON COURSE ASSIGNMENT

Cloud-Based College Event Registration System

A. Assignment Information

Particular	Details
Student Name	K.MAHESH BABU
Register Number	192525122
Department	AI&ML
Programme	B.Tech AIML
Course Code & Course Name	Cloud Computing
Academic Year / Batch	2025–2026
Faculty Name	Dr.K.Santhiya Lakshmi
Assignment Title	Cloud-Based College Event Registration System
Event Name	Tech Symposium
Participation Type	Individual
Maximum Marks	100
Contact details	8008553373 / maresh@example.com
Registration date	02-09-2026
Registration status	Confirmed

B. Assignment Problem / Challenge

Colleges regularly conduct events such as technical symposiums, cultural festivals, sports meets, workshops and hackathons, and these events attract registrations from a large number of students. Traditional registration methods, such as paper forms or manually maintained spreadsheets, make it difficult to consolidate entries, avoid duplicate or incomplete registrations, track participation category and generate quick, reliable statistics on turnout.

The challenge is to design and develop a simple cloud-based College Event Registration System using Zoho Creator. The system should allow students to register for events through an online

form, store the information in the cloud, organize registrations by event and participation category, and provide organizers with reports and a dashboard for monitoring and analysis.

The assignment requires students to apply cloud computing concepts, particularly SaaS, cloud data storage, cloud application development, reporting and data visualization, to a practical college-event-management problem.

C. Problem Statement

Design and develop a simple cloud-based College Event Registration System using Zoho Creator to demonstrate Software as a Service (SaaS). The system should allow students to register for college events by providing their basic details, selecting an event and choosing their participation category. The registration data should be stored and managed through the cloud.

The system should generate simple reports showing the number of registrations for each event and participation category, and the results should be analyzed to identify the most preferred events.

D. Requirements and Constraints

1. Functional Requirements

- Provide an online event registration form for students.
- Collect Student Name / ID and Department.
- Allow selection of Event Name such as Tech Symposium, Cultural Night, Sports Meet, Workshop on AI and Hackathon.
- Allow selection of Participation Category such as Individual and Team.
- Record contact details such as Email ID and Contact Number.
- Record the Date of registration.
- Store submitted registration records in the cloud.
- Provide a centralized registration report for organizers.
- Provide a dashboard showing the number of registrations per event and participation category.

- Identify and highlight the most preferred event(s) based on registration counts.

2. Non-Functional Requirements

- Usability: The interface should be simple and easy for students to understand.
- Reliability: Submitted registration information should be stored correctly without loss.
- Accessibility: The application should be accessible through internet-connected devices.
- Security: Registration and student information should be available only to authorized users.
- Scalability: The system should support an increasing number of event registrations.
- Maintainability: Organizers should be able to manage registration records without rebuilding the application.

3. Constraints

- The system depends on internet connectivity and cloud-platform availability.
- Features may depend on the selected Zoho Creator plan.
- Student registration information should be protected from unauthorized access.
- The project should remain simple enough for academic demonstration.
- The system should use realistic sample data for testing and dashboard validation.

E. Student Work

1. Problem Understanding and Formulation

Problem

Manual or informal event registration processes can make it difficult to maintain centralized records, avoid duplicate entries, track participation category and analyze which events students prefer the most.

Existing Process

Student → Paper/Spreadsheet Registration → Event Coordinator → Manual Record

Proposed Process

Student → Online Registration Form → Zoho Creator Cloud → Registration Report → Organizer Review → Dashboard

Expected Outcomes

- Digitize college event registration.
- Store registration information in the cloud.
- Organize registrations according to event, participation category and date.
- Enable organizers to track registration counts per event.
- Provide dashboard-based registration statistics.
- Demonstrate practical use of SaaS in a real-world application.

Sample Data Fields

Field	Example
Student Name / ID	Priya / 23AM045
Department	AI&ML
Event Name	Tech Symposium
Participation Category	Individual
Email ID	priya@example.com
Contact Number	9876543210
Date of Registration	02-09-2026

2. Application of Course Knowledge

Software as a Service (SaaS)

The application is accessed through the internet as a cloud-hosted service. Students and organizers can use the registration system without installing and maintaining a complete software system on their local computers.

Platform as a Service (PaaS)

Zoho Creator provides a cloud development environment and application-building tools for creating forms, reports, workflows and dashboard components without requiring the project team to manage the underlying physical infrastructure.

Cloud Accessibility

Students and authorized organizers can access the registration application from internet-connected devices, subject to service availability.

Data Management

Registration records are organized using Event Name, Participation Category, Department and Date of Registration so that organizers can search, filter and analyze the information.

3. Solution / Design / Methodology

Major Modules

Module	Description
Module 1 – Event Registration	Students enter their details, select an event and choose a participation category through the online form.
Module 2 – Cloud Registration Storage	Submitted registrations are stored in the Zoho Creator cloud application.
Module 3 – Event and Category Management	Organizers view registrations, filter records by event and participation category.
Module 4 – Dashboard and Reporting	The dashboard displays registration totals and charts by event and category, highlighting the most preferred events.

System Architecture

STUDENT



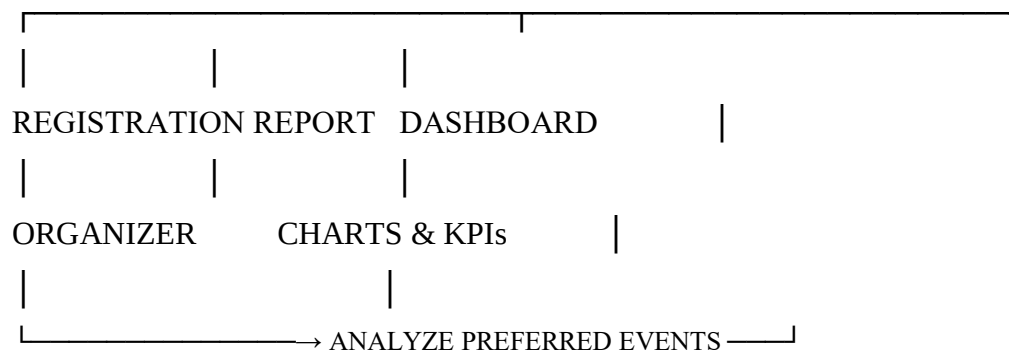
EVENT REGISTRATION FORM



ZOHO CREATOR CLOUD PLATFORM



CLOUD REGISTRATION DATA



Registration Status Workflow

New Registration → Confirmed → Attended

↘ Cancelled (if the student withdraws)

4. Use of Modern Tools

Zoho Creator will be used as the primary cloud application development platform. The project will use its form, report, dashboard and data-management capabilities to create the proposed system.

- Create the Event Registration Form.
- Create and manage registration data fields.

- Store registration information in the cloud.
- Create an All Registrations report.
- Create dashboard cards and charts.
- Filter and analyze registration records by event and category.
- Manage organizer/administrative access where applicable.

Evidence of tool usage should be included in the submission through screenshots of the application form, submitted records, registration report and dashboard.

5. Results and Validation

The system can be tested by entering representative registration records and checking whether the submitted information is stored and displayed correctly.

Sample Test Dataset

ID	Student Name	Event	Category	Status
R001	Priya	Tech Symposium	Individual	Confirmed
R002	Arjun	Hackathon	Team	Confirmed
R003	Sneha	Cultural Night	Individual	Confirmed
R004	Karthik	Sports Meet	Team	Confirmed
R005	Divya	Workshop on AI	Individual	Confirmed
R006	Rahul	Tech Symposium	Team	Confirmed
R007	Meena	Cultural Night	Individual	Cancelled
R008	Vignesh	Hackathon	Team	Confirmed
R009	Anitha	Sports Meet	Individual	Confirmed
R010	Suresh	Workshop on AI	Team	Confirmed

Dashboard Example

Category	Number
Tech Symposium registrations	8
Hackathon registrations	6
Cultural Night registrations	5
Sports Meet registrations	4
Workshop on AI registrations	6
Total registrations	29
Individual category registrations	15
Team category registrations	14
Most preferred event	Tech Symposium

Testing and Validation

Test Case	Expected Result
Submit valid registration	Registration should be stored successfully.
Select Tech Symposium event	Registration should be linked to Tech Symposium.
Select Individual category	Registration should display Individual category.
Select Team category	Registration should display Team category.
View registration report	Submitted registration should appear in the report.
Filter by event	Relevant event registrations should be displayed.
Open dashboard	Dashboard should show updated registration statistics.

6. Analysis and Engineering Decision

The proposed cloud-based system is preferable to a manual registration register because it centralizes registration information and provides easier tracking and analysis of event popularity.

Feature	Manual System	Cloud-Based System
Registration submission	Paper/in-person	Online
Data storage	Register/files	Cloud

Tracking	Difficult	Easy
Searching	Manual	Fast filtering/search
Statistics	Manual calculation	Dashboard
Accessibility	Limited	Internet-based
Event popularity analysis	Difficult	Automated via dashboard

Zoho Creator is selected because it provides a suitable cloud environment for developing a small application with forms, data management, reports and dashboards. This makes it appropriate for demonstrating cloud application development and SaaS concepts at an academic level.

7. Broader Considerations

- Sustainability: Digital event registration reduces paper usage.
- Society: The system provides a more organized communication channel between students and event organizers.
- Accessibility: Students can register for events without physically visiting an administrative office.
- Security and Privacy: Student and registration information should be accessible only to authorized users.
- Economics: Centralized digital management can reduce manual administrative effort and paper-related costs.
- Professional Responsibility: Registration records should be maintained accurately and should not be modified or disclosed without authorization.

8. Conclusion

The Cloud-Based College Event Registration System provides a practical approach to digitizing college event registration using cloud computing technology. Students can register for events through an online form, while organizers can store, organize, track and analyze registrations through a centralized cloud application.

The use of Zoho Creator demonstrates the practical application of Software as a Service (SaaS), cloud storage, online application access and cloud-based application development. The dashboard provides useful information about registration counts by event and participation category, supporting better monitoring and decision-making, including identification of the most preferred events.

The proposed system can reduce manual record keeping, improve registration visibility and provide a more organized method for managing college event participation.

9. References

Mell, P., & Grance, T. (2011). The NIST Definition of Cloud Computing. National Institute of Standards and Technology.

National Institute of Standards and Technology (NIST). Cloud Computing resources and guidance.

Zoho Corporation. Zoho Creator documentation and product resources.

Rittinghouse, J. W., & Ransome, J. F. (2017). Cloud Computing: Implementation, Management, and Security. CRC Press.

Erl, T., Puttini, R., & Mahmood, Z. (2013). Cloud Computing: Concepts, Technology & Architecture. Prentice Hall.

10. Student Reflection

Through this assignment, I learned how cloud computing concepts can be applied to solve a real-world problem. I understood the working of the SaaS model and learned how a cloud-based application can collect, store and analyze information.

I also learned how forms, reports and dashboards can be used to create a simple management system using a cloud platform.

If additional time and resources were available, the system could be improved by adding automatic notifications, administrator roles, complaint escalation, image upload for complaints and student feedback after complaint resolution.